

David Han

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EDUCATION

University of Southern California

Bachelor of Science in Computer Science (GPA: 3.77/4.0)

- **Minors:** AI Applications & Video Game Programming
- **Honors & Awards:** Presidential Scholar (2% of Students, Half Tuition), Viterbi Dean's List (6x)
- **Relevant Coursework:** Data Structures & Algorithms, Software Engineering, Computer/Embedded Systems, Operating Systems, Artificial Intelligence, Internetworking, Discrete Math, Linear Algebra & Differential Equations

Los Angeles, CA

May 2027 (Expected)

EXPERIENCE

Bloomberg L.P.

Software Engineering Intern

- Built a **Python** microservice speeding up the field migration process on a legacy **C++** analytics engine by **500%**
- Engineered a **React/TypeScript** dashboard surfacing traffic-weighted migration gaps across **1000+** unstructured fields
- Designed a **diff-analysis** pipeline with tolerance-banded deviation checks, catching **95%** of pre-release regressions
- Deployed using an internal PaaS, configuring custom auth and **Kubernetes** orchestration to handle **1M+** daily requests

New York, NY

June 2026 – Present

Activision Blizzard

Software Engineering Intern

- Spearheaded the development of a security dashboard for Call of Duty, processing millions of real-time anti-cheat detections
- Designed **5+** scalable backend microservices and data pipelines in **Go**, improving threat intelligence efficiency by **40%**
- Built modular frontend features in **TypeScript** + React, integrating Postgres-backed services to deliver valuable insights
- Implemented OAuth2.0 auth flows with Microsoft Entra and JWT caching, reducing authentication latency by **25%**

Sherman Oaks, CA

June 2025 – September 2025

Batchr

Founder & Lead Engineer

- Founded restaurant technology startup, building an ambient system for production planning and inventory management
- Architected a full-stack app with **React** + **Express.js**, implementing RESTful APIs and MongoDB schemas for products
- Secured paid partnership with Angela's Ice Cream, currently deploying scalable solutions and iterating across **8+** locations

Los Angeles, CA

January 2025 – August 2025

USC Viterbi School of Engineering

Undergraduate Tutor

- Tutored 50+ students in CSCI 103/170; covered graph theory, logic, asymptotics, probability, and **C++** programming basics
- Increased students' test scores by over **20%** through holistic review sessions, office hours, and self-curated problem sets

Los Angeles, CA

August 2024 – May 2025

Alarm.com

Software Engineering Intern

- Led development of a scheduling feature for home automation “scenes” using with **Ember** and **C#** in an Agile environment
- Engineered backend scheduler with **30+** optimized SQL queries & caching, boosting database performance by **15%**
- Transitioned **3+** legacy ASPX pages to modern JS frameworks, ensuring seamless integration and improved functionality

Lawrence, KS

June 2024 – August 2024

PROJECTS

Fleet Commander (*Python, Next.js, FastAPI, LangChain*) — [GitHub](#)

May 2025 – August 2025

- Developed multi-agent, zero-touch deployment pipeline enabling repositories to auto-analyze, test, and deploy via webhooks
- Engineered **FastAPI** backend with **LangChain** ReAct agents, WebSocket live status updates, and orchestrated CI/CD flow

Glance (*Electron, OpenAI, MongoDB*) — [GitHub](#)

August 2024 – May 2025

- Architected full-stack AI-powered documentation platform saving users **5+** hours weekly through automated insights
- Implemented distributed system for actionable insights using custom **machine learning** pipelines and **Tesseract OCR**

Let's Get Lyrical (*React, Java, Spring Boot, Gherkin*) — [GitHub](#)

January 2025 – May 2025

- Developed a **Java** backend with Spring Boot, implementing multi-threaded lyric scraping for sub-second word cloud creation
- Wrote extensive unit/integration tests using **Cucumber** and **JUnit**, covering authentication, comparison, and feature logic

TECHNICAL SKILLS

Languages: C++, Java, Python, Go, JavaScript, TypeScript, C#, SQL, Assembly, HTML/CSS, Lua, Bash, Swift

Tools/Frameworks: React, Electron, Express, PostgreSQL, MongoDB, Docker, Kubernetes, Cucumber, Git, Linux, AWS

LEADERSHIP & INVOLVEMENT

LavaLab F24 Cohort (Developer)

September 2024 - December 2024

- Built **Glance** with a PM and designer at USC's top product incubator (4% acceptance); led full-cycle product development
- Won **\$500** for Best Pitch at VC Demo Day, recognized for clarity of vision, product impact, and market potential

USC Theta Tau Professional Engineering Society (Media Chair)

December 2023 - December 2024

- Maintained and updated chapter website (**HTML/CSS**); created media content that increased rush engagement by **17%**